

Machine Learning for Oil and Gas Production Forecasting in the Permian Basin

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The National Energy Technology Laboratory (NETL) conducted a real-world machine-learning project to forecast daily oil, water, and natural-gas production from horizontal wells in the Midland Basin of West Texas. The study used a quasi-experimental design to compare models built with proprietary operator data against models using information more commonly available from public sources. Its central question was whether richer operational and geological data could materially improve production forecasting (Vikara et al., 2022).

Data Preprocessing

The original dataset included 318 horizontal wells completed in the Spraberry/Dean and Wolfcamp formations between 2015 and 2020. After records with critical missing information were removed, 235 wells remained for model development. The data included production histories, pressure measurements, well-completion information, spatial variables, and well-log characteristics. Missing data were handled according to operational context. When a well was shut in, missing production values could reasonably be represented as zero. Missing flow measurements during active production were interpolated, while missing pressure values were estimated using multivariate imputation. Numerical variables were normalized to a 0–1 range. Because well histories differed in length, sequences were padded to a common length so they could be processed by the neural network (Vikara et al., 2022).

This preprocessing approach aligns with recommended practice because missing values were not treated with a single generic method. Instead, the researchers considered the physical meaning of the data. A limitation, however, is that the report does not describe a formal statistical method for detecting outliers. In future work, I would add an explicit outlier-screening process and

combine statistical detection with engineering review so that unusual but legitimate production events are not mistakenly removed.

Model Selection and Validation

The researchers selected a Long Short-Term Memory (LSTM) neural network. This was appropriate because oil, water, and gas production are time-series variables, and LSTMs are designed to learn sequential relationships and changing patterns over time. Three model variants were created: a Proprietary Model, a Quasi-public Model, and a Public Model. The Proprietary Model included the richest operational, geological, pressure, and completion information, while the other models progressively reduced access to proprietary inputs. This allowed the researchers to isolate the value of additional data.

The wells were divided approximately 80% for training, 10% for validation, and 10% for holdout testing. Importantly, the split occurred at the well level, meaning the entire production history of a given well remained in only one partition. This strongly aligns with good validation practice because randomly splitting daily records from the same well could cause data leakage and artificially inflate test performance. Early stopping was also used to reduce overfitting (Vikara et al., 2022). One weakness is that model architecture and hyperparameters were refined largely through trial and error rather than systematic cross-validation or automated optimization. Although deep-learning models can be computationally expensive to tune, a more structured search process would strengthen confidence that the selected configuration was close to optimal.

Model Evaluation

Because the study predicted continuous production volumes, regression metrics were more appropriate than classification measures such as accuracy, precision, recall, and F1-score. NETL

evaluated performance using Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and R^2 (coefficient of determination).

The Proprietary Model achieved the strongest overall results, with average R^2 values of approximately 0.86 for oil and 0.79 for both water and gas production. It also produced substantially lower prediction errors than the Public Model. These findings indicate that high-frequency operational and geological information provided meaningful predictive value (Vikara et al., 2022). The study also showed that prediction quality weakened later in a well's producing life because fewer wells contained long production histories. This is an important reminder that strong average model performance can hide weaker performance in less-represented operating conditions.

Ethical Considerations

The primary ethical issue in this project involved privacy and proprietary data governance. NETL obtained operator-specific information that was not publicly available, and the operator's identity was protected. Secure data-sharing arrangements were therefore necessary to balance research value with commercial confidentiality.

Traditional fairness concerns involving protected demographic groups are less applicable because the model predicts well production rather than human outcomes. However, model bias remains relevant. The data came from a specific operator and geological region, so the model may not perform equally well for other operators, formations, or basins. External validation would therefore be necessary before using the model elsewhere.

Environmental responsibility should also be considered because production forecasts can influence drilling, water management, and extraction decisions. ML outputs should support engineering judgment and regulatory requirements rather than replace them.

Post-Deployment Monitoring and Reflection

The study does not provide a detailed post-deployment monitoring plan, which is its largest weakness relative to recommended ML practice. If deployed, I would continuously compare predicted and actual production using rolling RMSE and R^2 . I would also monitor important input variables for data drift and track performance by formation, well age, and completion type. Predetermined error thresholds should trigger engineering review and possible retraining.

The practices I would apply in future work are context-aware missing-data treatment, well-level data separation, appropriate regression metrics, and early stopping. I would improve the approach by adding formal outlier detection, more systematic hyperparameter tuning, external validation, and a documented monitoring and retraining process. Overall, this case demonstrates that reliable machine learning depends not only on predictive accuracy but also on sound preprocessing, realistic validation, responsible data governance, and continued verification after deployment.

AI Use Statement: I used ChatGPT to help organize and refine this report. I verified the technical information against the original NETL source and completed the final analysis and reflection myself.

References

Vikara, D., Bello, K., Wijaya, N., Warner, T., Sheriff, A., Hoffman, H., & Remson, D. (2022).

Evaluating the Impact of Proprietary Oil & Gas Data on Machine Learning Model

Performance Using a Quasi-experimental Analytical Approach (Report No. DOE/NETL-2022/3780). National Energy Technology Laboratory (NETL).

<https://doi.org/10.2172/1860128>



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Key Insights from a Real-World Case Study (NETL, 2022)

PROJECT OVERVIEW

NETL developed machine learning models to forecast daily oil, water, and natural gas production from horizontal wells in the Permian Basin. The study compared models built with proprietary operator data vs. publicly available data to determine the value of richer operational and geological information.



STUDY AT A GLANCE



318 horizontal wells (2015–2020)
235 wells used for modeling



Objective: Predict daily oil, water, and gas production



Design: Quasi-experimental comparison (Proprietary vs. Quasi-public vs. Public data)



Outcome: Proprietary data significantly improves prediction accuracy

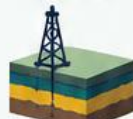
1. DATA PREPROCESSING

Handling Missing Data



- Shut-in periods represented as zero
- Active production gaps interpolated
- Missing pressure values imputed using multivariate methods

Feature Engineering & Geology



- Wellbore azimuth calculated
- Well logs clustered geographically
- Geological variability captured as features

Normalization



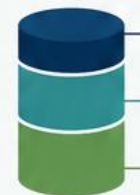
- All numerical features scaled to a 0–1 range

Sequence Alignment



- Sequences padded to common length (1,969 timesteps) with masking layer in model

Three Data Scenarios



- Proprietary** (richest data)
- Quasi-public** (moderate data)
- Public** (limited data)

2. MODELING & VALIDATION



Model: LSTM
(Long Short-Term Memory) Network

Ideal for capturing sequential patterns in production data.

Data Split (by Well)



- 80% Training
- 10% Validation
- 10% Testing



Well-level split prevents data leakage



Early stopping used to reduce overfitting



Hyperparameters refined through trial and error

3. MODEL PERFORMANCE (PROPRIETARY vs. PUBLIC)

Metric (Holdout Test Set)	Oil	Water	Gas
R^2 (Coefficient of Determination)	0.86	0.79	0.79
Daily RMSE Improvement	26% ↓	29% ↓	31% ↓
Cumulative Oil RMSE Improvement	40% ↓	—	—



Proprietary data provided substantially lower prediction errors and better explained variance across all three production streams.

4. ETHICAL CONSIDERATIONS



Privacy & Proprietary Data

Operator identity protected; secure data-sharing arrangements balance research value with commercial confidentiality.



Bias & Generalizability

Data from one operator and region may not generalize to other basins or operators. External validation is essential.



Environmental Responsibility

Forecasts influence drilling, water use, and extraction decisions; ML should support engineering judgment and regulation.

5. POST-DEPLOYMENT MONITORING PLAN (Recommended)



Monitor performance with rolling RMSE and R^2 against new field data



Detect data drift in key inputs (e.g., pressure, well status)



Retrain or fine-tune model when performance drops below threshold



Fallback to engineering decline curves during sensor outages

6. KEY INSIGHTS TO APPLY (and Avoid)



APPLY

- Use rich operational & high-frequency data (e.g., pressure) when available
- Incorporate geological variability
- Split data by well to prevent leakage
- Use appropriate regression metrics (RMSE, R^2)
- Implement monitoring and retraining plans



AVOID

- Relying only on public data when richer data can improve outcomes
- Random row-level splits for time-series data
- Ignoring operational context (shut-ins, restarts)
- Assuming one model fits all geological settings

MY REFLECTION



This case study was eye-opening. Public oil and gas data are limited, which constrains research and model performance. Industry collaboration is essential to access valuable operational and geological information. However, solutions are not one-size-fits-all—geological variation, operating practices, and data availability differ across operators and basins. Successful ML projects require the right data, domain expertise, and models tailored to specific needs.

